



BACKDROPS · SETUP

Open Scratch with a dark neon backdrop ready for the action



STEPS TO FOLLOW

1. Go to scratch.mit.edu → click "Create"
2. Right-click the cat sprite → Delete
3. Click "Choose a Backdrop" → search "Stars" or "Neon Tunnel"
4. (Optional) Backdrops tab → draw a bright neon horizon line
5. Sign in to save your project automatically

TIP

A dark backdrop makes neon-coloured sprites pop — same trick used by Geometry Dash and Tetris.

DID YOU KNOW?

The Scratch stage is 480×360 pixels. Y goes from -180 (bottom) to +180 (top). A horizon at $y=-100$ leaves perfect jump room.



SPRITES · COSTUMES

Paint a simple blue cube sprite with a tiny smiling face



STEPS TO FOLLOW

1. Hover "Choose a Sprite" → click **Paint** 
2. Costumes tab → pick Rectangle tool, bright blue fill, no outline
3. Drag a 50 × 50 square at the centre of the canvas
4. Brush tool → two black dots for eyes, tiny smile
5. Rename the sprite "**Cube**" in Sprite Info

TIP

Keep the cube small (~50 px) so jumps look snappy and spikes feel close to the action.

DID YOU KNOW?

Geometry Dash characters started as one-colour squares — drawn in 10 minutes. Simple shapes animate beautifully.



MOTION · EVENTS

On green flag, the cube starts at the left of the screen sitting on the ground



SCRATCH BLOCKS (CUBE SPRITE)

when  clicked

go to x: (-150) y: (-100)

set size to (80) %

TIP

Always reset position in the green-flag script. Players love restarting — your game must reset cleanly.

DID YOU KNOW?

"go to x y" is teleportation — instant, no animation. It's faster than "glide" when you just need to reset.



EVENTS · LOOPS

Press SPACE → the cube rises then falls back to the ground



ADD THIS NEW SCRIPT TO THE CUBE SPRITE

```
when [space▼] key pressed
  if <(y position) < (-95)> then
    repeat (10)
      change y by (12)
    end
  repeat (10)
    change y by (-12)
  end
end
```

TIP

The outer "if" stops mid-air double-jumping — "only jump if near the ground".

DID YOU KNOW?

This jump pauses the script. Step 5 upgrades it to a smarter version so spikes can still scroll while you jump.



VARIABLES · PHYSICS

Replace the simple jump with "y power" + gravity so spikes can scroll while you jump



REPLACE THE SPACE SCRIPT — TWO SCRIPTS NOW

```
when clicked
```

```
go to x: (-150) y: (-100)
```

```
set [y power▼] to (0)
```

```
forever
```

```
  change [y power▼] by (-1)
```

```
  change y by (y power)
```

```
  if <(y position) < (-100)> then
```

```
    set y to (-100)
```

```
    set [y power▼] to (0)
```

```
  end
```

```
end
```

```
when [space▼] key pressed
```

```
if <(y position) < (-95)> then
```

```
  set [y power▼] to (12)
```

💡 TIP

Positive y = up. y power = 12

launches up; gravity subtracts 1 each

frame so the leap arcs back down

naturally.

🧐 DID YOU KNOW?

Two scripts running at once is

"parallel programming". Real games

use threads or async — same idea.



SPRITES · PAINT EDITOR

Paint a small red triangle obstacle sprite



STEPS TO FOLLOW

1. Hover "Choose a Sprite" → click **Paint**
2. Costumes tab → Line tool, bright red, thickness 6
3. Draw 3 lines forming an upward triangle (~40 px wide)
4. Fill tool  → fill the inside red
5. Rename the sprite "**Spike**"

TIP

Anchor the spike's base at the bottom of the canvas — then "go to y: -110" plants it on the ground perfectly.

DID YOU KNOW?

Geometry Dash uses only a handful of shapes — spike, block, jump-pad — arranged in patterns. Limited art + clever rhythm = endless levels.



MOTION · LOOPS

**Spike slides from right to left, then loops back to the right edge
endlessly**



SCRATCH BLOCKS (SPIKE SPRITE)

when  clicked

go to x: (240) y: (-110)

forever

change x by (-6)

if <(x position) < (-240)> then

set x to (240)

TIP

"change x by (-6)" is the scroll speed.

Bigger negatives = faster. Step 10

makes this speed grow with the score!



DID YOU KNOW?

The character never moves forward —

the world moves past it. Subway

Surfers and Mario use the exact same

trick.



SENSING · CONTROL

If the cube touches a spike, the game stops everything instantly



ADD INSIDE CUBE'S FOREVER LOOP

```
if <touching [Spike▼]?> then
```

```
  say [Game Over!] for (2) secs
```

```
  stop [all▼]
```

```
end
```

TIP

"stop all" is the nuclear option — it freezes every script in every sprite.

Perfect for game over.

DID YOU KNOW?

In pro games this is a "hitbox check" — CPU compares rectangles 60 times a second to decide if you live or die.



VARIABLES · LOOPS

Score climbs by 1 every second you stay alive — survival is rewarded



NEW SCRIPT ON CUBE SPRITE

when  clicked

set [Score▼] to (0)

forever

wait (1) secs

change [Score▼] by (1)

TIP

Multiple "when  clicked" scripts are fine — Scratch runs them all in parallel, each handling one job.

DID YOU KNOW?

Time-based scoring is called "survival scoring". Subway Surfers, Temple Run, Fall Guys all use it.



VARIABLES · OPERATORS

Spike speeds up as Score climbs — automatic difficulty ramp



REPLACE "CHANGE X BY (-6)" ON SPIKE WITH:

```
change x by ( 0 - (6 + ((Score) / (5))) )
```

→ 0 minus (6 plus Score divided by 5)

→ as Score grows, the value gets more negative

→ more negative x = faster left scroll!

TIP

Start gentle (6 + Score/5). Too easy?

Try Score/3. Too hard? Try Score/10.

Tune to taste!

DID YOU KNOW?

This is a "difficulty curve". Tetris

speeds up every 10 lines, Pac-Man

ghosts roam faster each maze. Same

trick everywhere.



SPRITES · COSTUMES

Add a TopSpike hanging from the ceiling — don't jump when it's overhead!



DUPLICATE SPIKE → TOPSPIKE (FLIPPED, Y=110)

```
when  clicked
  go to x: (120) y: (110)
  forever
    change x by ( 0 - (6 + ((Score) / (5))) )
    if <(x position) < (-240)> then
      set x to (240)
    end
  end
end

- update Cube's collision -

if <<touching [Spike▼]?> or <touching [TopSpike▼]?>> then
  stop [all▼]
```

 TIP

Set TopSpike's starting x DIFFERENT from the ground Spike (120 vs 240) — otherwise both arrive together and the game is unfair.

 DID YOU KNOW?

Geometry Dash mixes ground and ceiling spikes in "fly-through-gap" sections. "Two threats, pick one" is core action-game design.



LOOKS · OPERATORS

Paint 3 spike costumes; each respawn picks a random look



SPIKE SPRITE — PAINT 3 COSTUMES FIRST

1. Costumes tab → right-click → duplicate (×2 more)
2. spike2: change colour to **purple**, add zig-zags
3. spike3: change colour to **gold**, replace with lightning

— *add inside Spike's respawn check* —

```
if <(x position) < (-240)> then
```

```
  set x to (240)
```

```
  switch costume to (pick random (1) to (3))
```

```
end
```

TIP

Costume numbers are 1-based. "pick random 1 to 3" matches your 3 painted spikes exactly.

DID YOU KNOW?

Game artists call this "asset variation" — same shape, many skins. Subway Surfers and Crossy Road use a tiny art library this way.



LOOKS · BACKDROPS

3 backdrops cycle every 10 seconds — your world keeps changing



STAGE SCRIPT (CLICK STAGE THUMBNAIL)

First: add 2 more backdrops (Forest, Volcano)

when  clicked

switch backdrop to (Stars▼)

forever

wait (10) secs

next backdrop

end

 TIP

Want scenes tied to score? Replace

"wait (10) secs" with "wait until

<(Score) mod (5) = 0>" — scene

advances every 5 points.



DID YOU KNOW?

Big games call this "biome cycling".

Minecraft has biomes, Mario Galaxy

has galaxies — variety keeps players

hooked for hours.